

120 Days of Quantitative Finance

Month 5: Advanced Quant Systems

Week 17: Volatility & Risk

Day 114: Volatility Surface and Implied Volatility Smile

1 Introduction

In practice, volatility is not constant across options. Instead, implied volatility varies with strike price and time to maturity.

This leads to the concept of the volatility smile and volatility surface.

2 Implied Volatility

Implied volatility is the value of σ that matches the observed market price of an option:

$$C_{market} = C_{BS}(S, K, T, r, \sigma_{imp})$$

3 Volatility Smile

For a fixed maturity, implied volatility varies with strike:

$$\sigma_{imp} = f(K)$$

Typically:

- Higher volatility for deep in-the-money and out-of-the-money options

- Lower volatility near at-the-money

4 Volatility Surface

Extending across maturities:

$$\sigma_{imp} = f(K, T)$$

This forms a 3D surface.

5 Stylized Facts

- Equity markets exhibit volatility skew (downside risk priced higher)
- Volatility increases during market stress
- Smile shape varies across assets

6 Why Black-Scholes Fails

Black-Scholes assumes:

$$\sigma = \text{constant}$$

But market prices imply:

$$\sigma = f(K, T)$$

7 Connection to Stochastic Volatility

Models like Heston generate volatility smiles naturally:

- stochastic volatility
- correlation between returns and volatility

8 Financial Interpretation

- Implied volatility reflects market expectations and risk premiums
- Smile captures tail risk and demand for protection

9 Applications

- option pricing and calibration
- volatility trading
- risk management

10 Conclusion

The volatility surface provides a more complete picture of market expectations than a single volatility value, highlighting the limitations of constant volatility models.